Question ID: 7812801f

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

The fynbos shrubland is a diverse habitat found only in South Africa. It is adjacent to the Afro-temperate forest, with almost no transition space between the two distinct habitats. Plants in the fynbos have uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil. Ecologists transplanted tree seedlings from the forest into plots in the fynbos. Seedlings in plots isolated from the roots of fynbos plants exhibited a growth rate five times greater than that of the seedlings in plots in close proximity to the roots of fynbos plants.

Based on the text, what role do fynbos roots most likely have in maintaining the border between the fynbos shrubland and the Afro-temperate forest habitats?

Answer

A. Fynbos roots damage the root systems of forest plants, leaving those plants unable to acquire sufficient nutrients.

B. Fynbos roots extend close enough to the forest plants' roots that they constitute a physical barrier that forest plants' roots cannot pass.

C. The root systems of fynbos plants allow the plants to take in so many soil nutrients that forest plants are prevented from flourishing in the fynbos.

D. The root systems of fynbos plants enhance the soil immediately surrounding the plants, allowing them to thrive in an otherwise harsh habitat.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a conclusion about the role of fynbos roots in maintaining the border with the Afro-temperate forest that can be reasonably inferred from the text. The text states that plants in the fynbos have "uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil." The text then describes an experiment where forest tree seedlings grown in plots isolated from fynbos roots grew at a rate five times greater than seedlings in areas near fynbos roots. This strongly suggests that fynbos roots are depleting soil nutrients to such an extent that forest plants can't get enough nutrients from that soil to flourish, thereby hindering the formation of any significant "transition space" between the habitats and helping keep the border between them sharp.

Choice A is incorrect because there is no evidence in the text that fynbos roots physically damage the root systems of forest plants. Instead, the text explicitly describes fynbos plants as having "uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil." This description establishes that fynbos roots are specialized for efficient nutrient absorption. The experiment then confirms this function of the roots by showing that forest seedlings "isolated from the roots of fynbos plants exhibited a growth rate five times greater" than those near fynbos roots. This result demonstrates that when forest seedlings don't have to compete with the nutrient-absorbing fynbos roots, they grow much better. Choice B is incorrect. Although the text does indicate that the fynbos and the Afro-temperate forest are distinct from, and adjacent to, each other and that there is "almost no transition space" (area of mixed fynbos and forest plants), the experiment shows that forest plants can grow in soil occupied by fynbos roots, just not very well. So, the idea that the fynbos roots physically obstruct forest plants is unsupported by the text. Choice D is incorrect because it directly contradicts the text's claim that fynbos roots absorb nutrients from, and therefore deplete, the soil they're in, and furthermore, nothing in the text suggests that the habitat of the fynbos is harsh in general.